Topic 2 Atoms, Elements and Compounds

Cambridge IGCSE Chemistry 0620 · Extended · Sections match the PMT / SaveMyExams topic numbers (2.1–2.7)

2.1 Elements, compounds and mixtures

	Element	Compound	Mixture
What it is	only **one type of atom**	two or more elements **chemically bonded** together	two or more substances **not chemically combined**
Examples	Fe, He, O_2, Cl_2	H_2O, NaCl, CO_2	air, sea water, crude oil
Proportions	—	**fixed** (shown by the formula)	can vary
Properties	—	**different** from the elements in it	same as the substances in it
Separate by	can't be split into anything simpler	only a **chemical reaction**	**physical methods** (filtering, distillation, chromatography…)

Example that makes it click: iron filings + sulfur powder is a *mixture* — a magnet pulls the iron out. Heat them and they react to make iron(II) sulfide, FeS, a *compound*: it isn't magnetic and you can't get the iron back with a magnet.

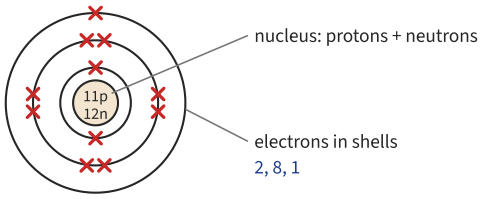
> ***Watch out***
> O_2 and Cl_2 are **elements** even though they are molecules — only one type of atom. "Molecule" and "compound" don't mean the same thing.

2.2 Atomic structure and the Periodic Table

An atom has a tiny central **nucleus** containing **protons** and **neutrons**, surrounded by **electrons** in **shells**.

Particle	Relative charge	Relative mass	Where
proton	+1	1	nucleus
neutron	0	1	nucleus
electron	−1	1/1840	shells

Atoms are neutral: **number of electrons = number of protons**.



Sodium atom, $^{23}_{11}Na$

> ***Definitions***
> **Proton number** (atomic number) – the number of protons in the nucleus of an atom.
> **Mass number** (nucleon number) – the total number of protons and neutrons in the nucleus.

Reading an atom symbol

In $^{23}_{11}Na$ the **top** number is the mass number and the **bottom** is the proton number.

protons = **11** · electrons = **11** · neutrons = 23 − 11 = **12**

Neutrons = mass number − proton number

Electronic configuration (elements 1–20)

Shells fill in order: **2** in the 1st, **8** in the 2nd, **8** in the 3rd, then the 4th (K, Ca).

- outer-shell electrons = **group number** (Groups I–VII)
- number of occupied shells = **period number**
- Group VIII (noble gases): **full outer shell**

Atom / ion	H	C	O	Na	Cl	Ar	Ca	Na^+	O^{2-}	Cl^-	Ca^{2+}
Proton number	1	6	8	11	17	18	20	11	8	17	20
Configuration	1	2,4	2,6	2,8,1	2,8,7	2,8,8	2,8,8,2	2,8	2,8	2,8,8	2,8,8

> ***Watch out***
> Work out electrons from the **proton number**, never the mass number. For ions: a **positive** charge means electrons **removed**, a **negative** charge means electrons **added** (O^{2-} has 8 + 2 = 10 electrons).

2.3 Isotopes

> ***Definition***
> **Isotopes** – different atoms of the same element that have the same number of protons but different numbers of neutrons.

	$^{35}_{17}Cl$	$^{37}_{17}Cl$	$^{35}_{17}Cl^-$
protons	17	17	17
neutrons	18	20	18
electrons	17	17	18

Isotopes have the same chemical properties because they have the same number of electrons, so the same electronic configuration. (Chemistry is decided by electrons, not neutrons.)

Relative atomic mass, A_r

A_r = the average mass of the isotopes of an element, compared to 1/12 of the mass of a carbon-12 atom. It's an average weighted by how common each isotope is (its **abundance**).

> ***Calculation***
> $A_r = (\text{mass}_1 \times \%_1 + \text{mass}_2 \times \%_2) \div 100$ e.g. chlorine: 75% Cl-35 and 25% Cl-37
> $A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 = \mathbf{35.5}$
> Sense check: the answer must lie between the isotope masses, and nearer the more common one.

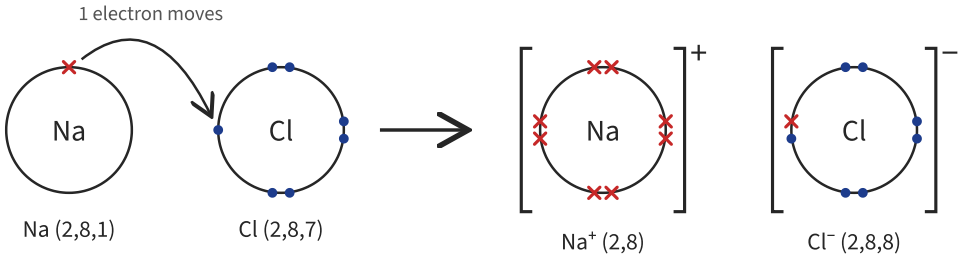
2.4 Ions and ionic bonds

Atoms bond to get a full outer shell. Three ways: **ionic** (metal + non-metal, electrons transferred), **covalent** (non-metals, electrons shared), **metallic** (metal atoms, outer electrons delocalised).

> ***Definitions***
> **Ion** – an atom or group of atoms with a charge, because it has lost or gained electrons.
> **Ionic bond** – the strong electrostatic attraction between oppositely charged ions.

- Metals **lose** electrons → positive ions
- Non-metals **gain** electrons → negative ions
- Charge = electrons lost or gained to reach a full shell

Group	I	II	III	V	VI	VII
Ion charge	+1	+2	+3	−3	−2	−1



Sodium + chlorine → sodium chloride. Na gives its 1 outer electron to Cl; both ions now have a full outer shell.

> ***Drawing ionic dot-and-cross diagrams***
> Dots for one atom's electrons, crosses for the other's · outer shell only unless asked · square brackets round each ion with the charge outside, top right · check every ion has a full outer shell.

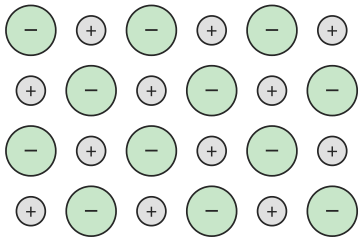
Other examples: Mg (2,8,2) gives 2 electrons to O (2,6) → Mg^{2+} O^{2-} (MgO). Mg gives 1 electron to *each* of two Cl atoms → Mg^{2+} $2Cl^-$ ($MgCl_2$).

Working out ionic formulas

The charges must add up to zero.

$Mg^{2+} + Cl^-$	need 2 Cl^-	→ **$MgCl_2$**
$Al^{3+} + O^{2-}$	2 × (+3) = +6, 3 × (−2) = −6	→ **Al_2O_3**

Shortcut: swap the charge numbers and drop the signs, then simplify (Mg^{2+} O^{2-} → Mg_2O_2 → MgO).



Cl^- (large) Na^+ (small)

NaCl lattice (one layer)

Giant ionic lattice

A regular arrangement of alternating positive and negative ions in 3D. There are no molecules — NaCl just gives the **ratio** of ions (1 : 1).

Property of ionic compounds	Why
High melting and boiling points	Giant lattice with many **strong electrostatic attractions between oppositely charged ions**; a lot of energy is needed to overcome them.
Conduct when molten or in aqueous solution	The **ions are free to move** and carry the charge.
Don't conduct when solid	Ions are held in fixed positions in the lattice and can't move.

Watch out

In ionic compounds it's the **ions** that move, not electrons · NaCl is not a "molecule" and has no "intermolecular forces" · ionic bonds are strong.

2.5 Simple molecules and covalent bonds

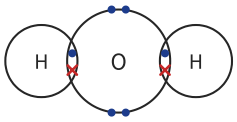
Definition

Covalent bond – a pair of electrons shared between two atoms, giving each atom a noble gas electronic configuration (full outer shell). Forms between non-metals.

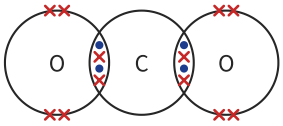
Single bond = 1 shared pair (–) · **double** = 2 pairs (=) · **triple** = 3 pairs (≡).
Electrons not in bonds sit in pairs: **lone pairs**.

Bonds an atom makes = electrons it needs to fill its outer shell:

Atom	H	Cl	O	N	C
Bonds	1	1	2	3	4



H_2O: 2 single bonds, 2 lone pairs on O



CO_2: 2 double bonds

Drawing covalent dot-and-cross diagrams

Overlap the outer shells of bonded atoms · one dot + one cross in the overlap for each bond · leftover electrons in pairs · check every atom has 8 in its outer shell (H has 2), and each atom still shows its own number of outer electrons (O = 6, N = 5, C = 4, H = 1).

Molecules you must be able to draw

Molecule	Displayed	Shared pairs	Lone pairs	Molecule	Displayed	Shared pairs	Lone pairs
H_2	H–H	1	0	**O_2**	O=O	2	2 on each O
Cl_2	Cl–Cl	1	3 on each Cl	**N_2**	N≡N	3	1 on each N
HCl	H–Cl	1	3 on Cl	**CO_2**	O=C=O	4	2 on each O
H_2O	H–O–H	2	2 on O	**C_2H_4**	$H_2C=CH_2$	6	0
NH_3	3 × N–H	3	1 on N	**CH_3OH**	$H_3C–O–H$	5	2 on O
CH_4	4 × C–H	4	0				

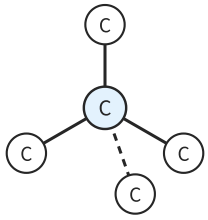
Property of simple molecular substances	Why
Low melting and boiling points (often gases or liquids at room temperature)	Only **weak intermolecular forces** between molecules; little energy is needed to overcome them. The covalent bonds *inside* the molecules are strong and do **not** break.
Poor electrical conductivity	Molecules have no overall charge: **no ions or free electrons** to carry charge.

Watch out — most common lost mark in this topic

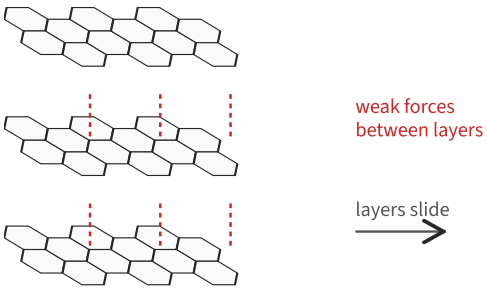
"When water boils, the covalent bonds break" is **wrong**: the intermolecular forces are overcome and steam is still H_2O. Covalent bonds are strong; the forces *between* molecules are weak.

2.6 Giant covalent structures

A huge network of atoms all joined by strong covalent bonds — no separate molecules. So: **very high melting points** (many strong covalent bonds to break), insoluble, and they don't conduct — except graphite.



Diamond: each C bonded to 4 others



Graphite: layers of hexagons, each C bonded to 3 others

	Diamond	Graphite
Structure	each C bonded to **4** others; rigid 3D (tetrahedral) network	each C bonded to **3** others in flat layers of hexagons; weak forces between layers
Free electrons	none — all 4 outer electrons used in bonds	1 **delocalised electron** per C atom
Properties	very hard · no conductivity · very high m.p.	soft, slippery · conducts · very high m.p.
Uses	**cutting tools** (drill tips)	**lubricant**, **electrodes**

Link structure to use (common 2–3 mark question)

Diamond → cutting: each C bonded to 4 others by strong covalent bonds in a rigid structure, so it is very hard.

Graphite → lubricant: weak forces between the layers, so the layers slide over each other.

Graphite → electrodes: delocalised electrons can move through the structure, so it conducts; very high melting point too.

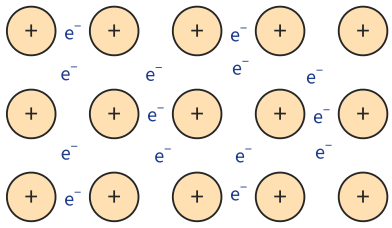
Silicon(IV) oxide, SiO_2 (sand): giant covalent — each **Si bonded to 4 O**, each **O bonded to 2 Si** (hence the 1 : 2 formula). Similar to diamond: hard, very high melting point, doesn't conduct — because both are giant networks of strong covalent bonds with no free electrons.

2.7 Metallic bonding

Definition

Metallic bonding – the electrostatic attraction between the positive ions in a giant metallic lattice and a "sea" of delocalised electrons.

Property	Why
Good electrical conductivity	**Delocalised electrons can move** through the structure and carry charge.
Malleable and ductile	**Layers of positive ions can slide** over each other; the sea of electrons keeps holding them together, so the metal bends instead of breaking.



Positive ions in a sea of delocalised electrons

★ Putting it together

Structure	Examples	Melting point	Conducts as solid?	Conducts molten / dissolved?
Giant ionic	NaCl, MgO	high	no	yes (ions move)
Simple molecular	H_2O, CO_2, CH_4	low	no	no
Giant covalent	diamond, graphite, SiO_2	very high	no (graphite yes)	— (insoluble)
Giant metallic	Cu, Fe, Al	usually high	yes (electrons move)	yes

Identify the structure from data

1. Conducts as a solid? Yes → metal (or graphite)
2. Conducts only when molten / dissolved? → giant ionic
3. Neither: **low** m.p. → simple molecular; **very high** m.p. → giant covalent

"Explain the melting point" in 3 steps

1. Name the structure. **2.** Name what must be overcome (attraction between oppositely charged ions / weak intermolecular forces / many strong covalent bonds / attraction between positive ions and delocalised electrons). **3.** "so a lot of / little energy is needed."

Self-check before the topic questions

- ☐ Tell elements, compounds and mixtures apart, with examples
- ☐ Give the relative charge, relative mass and position of protons, neutrons and electrons
- ☐ Work out protons, neutrons and electrons from a symbol like $^{23}_{11}Na$ or $^{35}_{17}Cl^-$
- ☐ Write electronic configurations for elements 1–20 and their ions
- ☐ Define isotopes and explain why they have the same chemical properties
- ☐ Calculate A_r from isotope masses and abundances
- ☐ Draw dot-and-cross diagrams for NaCl, MgO, $MgCl_2$; work out ionic formulas
- ☐ Draw H_2, Cl_2, HCl, H_2O, NH_3, CH_4, O_2, N_2, CO_2, C_2H_4, CH_3OH
- ☐ Explain melting point and conductivity for all four structures
- ☐ Compare diamond, graphite and SiO_2 and link to uses